

Task 2 Hazelcast https://github.com/hhafiya/software_architecture.git (hazelcast branch)

Крок 1. Встановити та налаштувати Hazelcast

Я використовувала docker образ `hazelcast/hazelcast:5.6`. Для клієнтської частини python застосунку я додала бібліотеку `hazelcast-python-client` через команду `uv add` до `pyproject.toml` та `uv.lock` для усіх трьох мікросервісів, що знаходяться в окремих піддиректоріях.

Крок 2. Конфігурація та запуск кластера з 3-х нод

Тут я реалізувала варіант окремих застосувань (Client-Server) через докер контейнери.

Пояснення налаштувань:

- Кожна нода має однаковий параметр `HZ_CLUSTER_NAME=dev`. За ним вони розуміють, що належать до однієї групи.
- Використовую TCP/IP Join `HZ_NETWORK_JOIN_TCP_IP_ENABLED=true`
- Вказую у конфігурації імена всіх трьох сервісів `HZ_NETWORK_JOIN_TCP_IP_MEMBERS=hz-node-1,hz-node-2,hz-node-3`
- Кожна нода повідомляє кластеру свою адресу через `HZ_NETWORK_PUBLIC_ADDRESS`. Це для того, щоб Management center та Python клієнт знали куди саме звертатися за даними.
- При запуску першою стартує `hz-node-1` і чекає на інших. `hz-node-2` бачить `hz-node-1` і з'єднується з нею. `hz-node-3` з'єднується з будь-якою з них.
- Порти назовні (з хост машини) 5701, 5702, 5703. Всередині мережі Docker використовується стандартний порт для Hazelcast 5701, бо ноди звертаються одне до одного за допомогою імен сервісів та внутрішнього порту.
- Також я одразу налаштувала для моніторингу стану кластера та візуалізації розподілу даних Management Center, який автоматично підключається до вузлів через параметр `HZ_MC_DEFAULT_CLUSTER_MEMBERS`. Його порт ззовні 8081, всередині докеру використовується 8080.
- В третьому кроці я змінила таку конфігурацію на `docker-compose + hazelcast.yml`. Прибрала `HZ_NETWORK_PUBLIC_ADDRESS` для більш стандартного підходу до роботи в Docker-мережі.

Крок 3. Демонстрація роботи Distributed Map

Для цього завдання я написала невеликий скрипт на Python `task3.py`, що знаходиться всередині counter сервісу. Він записує 1000 значень у mapу.

Спершу стандартно: `docker-compose up -d --build`

Для запуску я використовую таку команду: `docker-compose exec counter-service uv run python task3.py`. Вона запускає цей скрипт в уже запущеному контейнері.

У management center я заходжу через браузер за адресою `localhost:8081`.

1. Розподіл між трьома нодами:

Member ^	Entries	Gets	Puts	Removals	Sets	Enti
172.18.0.4:5701	340	0	3,247	0	0	
172.18.0.5:5701	324	0	324	0	0	
172.18.0.7:5701	336	0	336	0	0	
TOTAL	1,000	0	3,907	0	0	

```
hz-node-1-1 | Members (size:3, ver:19) {
hz-node-1-1 |   Member [172.18.0.4]:5701 - 5e74a783-9d11-4f7e-8436-7cb783a7119c
hz-node-1-1 |   Member [172.18.0.7]:5701 - a9f15f2d-d874-4a94-80bc-d98089526f2f
hz-node-1-1 |   Member [172.18.0.5]:5701 - 66aa9bd9-f8ee-472b-b5df-fd61f6a2d396 this
hz-node-1-1 | }
hz-node-1-1 | 2026-02-25 12:16:07,434 [ INFO] [main] [c.h.j.i.JobCoordinationService]: [172.18.0.5]:5701 [dev] [5.6.0] Jet started scanning for jobs
hz-node-1-1 | 2026-02-25 12:16:07,440 [ INFO] [main] [c.h.c.LifecycleService]: [172.18.0.5]:5701 [dev] [5.6.0] [172.18.0.5]:5701 is STARTED
hz-node-2-1 | 2026-02-25 12:16:07,524 [ INFO] [hz.focused_tharp_migration] [c.h.i.p.i.MigrationManagerImpl]: [172.18.0.4]:5701 [dev] [5.6.0] Repartitionin
g cluster data. Migration tasks count: 542
hz-node-2-1 | 2026-02-25 12:16:07,986 [ INFO] [hz.focused_tharp_migration] [c.h.i.p.i.MigrationManagerImpl]: [172.18.0.4]:5701 [dev] [5.6.0] All migration
tasks have been completed. (repartitionTime=Wed Feb 25 12:16:07 GMT 2026, plannedMigrations=542, completedMigrations=542, remainingMigrations=0, totalComp
letedMigrations=2786)
hz-node-1-1 | 2026-02-25 12:16:08,334 [ INFO] [hz.floppyant_galileo_priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [172.18
.0.5]:5701 [dev] [5.6.0] Received auth from Connection[id=3, /172.18.0.5:5701->/172.18.0.5:39745, qualifier=null, endpoint=[172.18.0.5]:39745, remoteUide=b
7ce1898-f6e2-4704-8097-73cf6eb346ec, alive=true, connectionType=MCJW, planeIndex=1], successfully authenticated, clientUide: b7ce1898-f6e2-4704-8097-73cf
6eb346ec, client name: MC-Client-dev, client version: 5.5.2
hz-node-1-1 | 2026-02-25 12:16:21,311 [ INFO] [hz.floppyant_galileo_priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [172.18
.0.5]:5701 [dev] [5.6.0] Received auth from Connection[id=4, /172.18.0.5:5701->/172.18.0.3:38978, qualifier=null, endpoint=[172.18.0.3]:38978, remoteUide=
55840bc-c70d-4663-b1a0-d0e021fc2d14, alive=true, connectionType=PH, planeIndex=1], successfully authenticated, clientUide: e55840bc-c70d-4663-b1a0-d0e021
fc2d14, client name: hz.client-0, client version: 5.6.0
hz-node-1-1 | 2026-02-25 12:16:21,314 [ INFO] [hz.focused_tharp_priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [172.18.0.4]
:5701 [dev] [5.6.0] Received auth from Connection[id=20, /172.18.0.4:5701->/172.18.0.3:49040, qualifier=null, endpoint=[172.18.0.3]:49040, remoteUide=e5584
0bc-c70d-4663-b1a0-d0e021fc2d14, alive=true, connectionType=PH, planeIndex=1], successfully authenticated, clientUide: e55840bc-c70d-4663-b1a0-d0e021fc2d
14, client name: hz.client-0, client version: 5.6.0
hz-node-3-1 | 2026-02-25 12:16:21,316 [ INFO] [hz.blissful_hugle.generic-operation.thread-2] [c.h.c.i.p.t.AuthenticationMessageTask]: [172.18.0.7]:5701 [d
ev] [5.6.0] Received auth from Connection[id=4, /172.18.0.7:5701->/172.18.0.3:36934, qualifier=null, endpoint=[172.18.0.3]:36934, remoteUide=e55840bc-c70d-
4663-b1a0-d0e021fc2d14, alive=true, connectionType=PH, planeIndex=1], successfully authenticated, clientUide: e55840bc-c70d-4663-b1a0-d0e021fc2d14, clien
t name: hz.client-0, client version: 5.6.0
```

2. Далі я вимкнула одну ноду після запуску скрипта через:
docker-compose stop hz-node-3

```
(facade-service) ira@MacBook-Air-Ira software_architecture % docker-compose stop hz-node-3
[+] stop 1/1
✓ Container software_architecture-hz-node-3-1 Stopped
```

Member	Entries	Gets	Puts	Removals	Sets	Ents
172.18.0.4:5701	482	0	1,904	0	0	
172.18.0.5:5701	518	0	2,146	0	0	
TOTAL	1,000	0	4,050	0	0	

```
hz-node-3-1 exited with code 143
hz-node-2-1 | 2026-02-25 12:19:46,789 [ INFO] [hz.focused_tharp.cached.thread-8] [c.h.i.s.t.TcpServerConnector]: [172.18.0.4]:5701 [dev] [5.6.0] Could not
connect to: /172.18.0.7:5701, Reason: IOException[Connect timed out to address /172.18.0.7:5701]
hz-node-1-1 | 2026-02-25 12:19:46,792 [ INFO] [hz.flamboyant_galileo.cached.thread-8] [c.h.i.s.t.TcpServerConnector]: [172.18.0.5]:5701 [dev] [5.6.0] Cou
ld not connect to: /172.18.0.7:5701, Reason: IOException[Connect timed out to address /172.18.0.7:5701]
hz-node-2-1 | 2026-02-25 12:19:46,892 [ INFO] [hz.focused_tharp.cached.thread-8] [c.h.i.s.t.TcpServerConnector]: [172.18.0.4]:5701 [dev] [5.6.0] Connectin
g to /172.18.0.7:5701, timeout: 10000, bind-any: true
hz-node-2-1 | 2026-02-25 12:19:56,894 [ INFO] [hz.focused_tharp.cached.thread-8] [c.h.i.s.t.TcpServerConnector]: [172.18.0.4]:5701 [dev] [5.6.0] Could not
connect to: /172.18.0.7:5701, Reason: IOException[Connect timed out to address /172.18.0.7:5701]
hz-node-2-1 | 2026-02-25 12:19:56,895 [ WARN] [hz.focused_tharp.cached.thread-8] [c.h.i.s.t.TcpServerConnectionErrorHandler]: [172.18.0.4]:5701 [dev] [5.6
.0] Removing connection to endpoint [172.18.0.7:5701] Cause => java.io.IOException[Connect timed out to address /172.18.0.7:5701], Error-Count: 5
hz-node-2-1 | 2026-02-25 12:19:56,895 [ INFO] [hz.focused_tharp.cached.thread-8] [c.h.i.c.i.MembershipManager]: [172.18.0.4]:5701 [dev] [5.6.0] Removing M
ember [172.18.0.7:5701 - 66aa9bd9-f8ee-472b-b5df-fd61f6a2d396]
hz-node-2-1 | 2026-02-25 12:19:56,896 [ INFO] [hz.focused_tharp.cached.thread-8] [c.h.i.p.i.PartitionStateManagerImpl]: [172.18.0.4]:5701 [dev] [5.6.0] St
oring snapshot of partition assignments while removing UUID a9f15f2d-0874-4a94-a0bc-d988895620f2
hz-node-2-1 | 2026-02-25 12:19:56,897 [ INFO] [hz.focused_tharp.cached.thread-8] [c.h.i.c.ClusterService]: [172.18.0.4]:5701 [dev] [5.6.0]
hz-node-2-1 | Members (size:2, ver:20) [
hz-node-2-1 | Member [172.18.0.4]:5701 - 5e74a703-9d11-4f7e-8436-7cb783a7119c this
hz-node-2-1 | Member [172.18.0.5]:5701 - 66aa9bd9-f8ee-472b-b5df-fd61f6a2d396
hz-node-2-1 | ]
hz-node-2-1 | 2026-02-25 12:19:56,897 [ INFO] [hz.focused_tharp.cached.thread-11] [c.h.t.TransactionManagerService]: [172.18.0.4]:5701 [dev] [5.6.0] Commi
tting/rolling-back live transactions of [172.18.0.7]:5701, UUID: a9f15f2d-0874-4a94-a0bc-d988895620f2
hz-node-1-1 | 2026-02-25 12:19:56,901 [ INFO] [hz.flamboyant_galileo.priority-generic-operation.thread-8] [c.h.i.p.i.PartitionStateManagerImpl]: [172.18.0
.5]:5701 [dev] [5.6.0] Storing snapshot of partition assignments while removing UUID a9f15f2d-0874-4a94-a0bc-d988895620f2
hz-node-1-1 | 2026-02-25 12:19:56,899 [ INFO] [MC-Client-dev.event-2] [c.h.w.s.MCClientManager]: Member removed: Member [172.18.0.7]:5701 - a9f15f2d-0874-
4a94-a0bc-d988895620f2
hz-node-1-1 | 2026-02-25 12:19:56,910 [ INFO] [hz.flamboyant_galileo.priority-generic-operation.thread-8] [c.h.i.c.ClusterService]: [172.18.0.5]:5701 [dev
] [5.6.0]
hz-node-1-1 | Members (size:2, ver:20) [
hz-node-1-1 | Member [172.18.0.4]:5701 - 5e74a703-9d11-4f7e-8436-7cb783a7119c
hz-node-1-1 | Member [172.18.0.5]:5701 - 66aa9bd9-f8ee-472b-b5df-fd61f6a2d396 this
hz-node-1-1 | ]
hz-node-1-1 | 2026-02-25 12:19:56,921 [ INFO] [hz.flamboyant_galileo.cached.thread-1] [c.h.t.TransactionManagerService]: [172.18.0.5]:5701 [dev] [5.6.0] C
ommitting/rolling-back live transactions of [172.18.0.7]:5701, UUID: a9f15f2d-0874-4a94-a0bc-d988895620f2
hz-node-1-1 | 2026-02-25 12:19:56,940 [ INFO] [hz.focused_tharp.migration] [c.h.i.p.i.MigrationManagerImpl]: [172.18.0.4]:5701 [dev] [5.6.0] Successfully
promoted 98 backups on 2 members in 43 ms
hz-node-2-1 | 2026-02-25 12:19:56,942 [ INFO] [hz.focused_tharp.migration] [c.h.i.p.i.MigrationManagerImpl]: [172.18.0.4]:5701 [dev] [5.6.0] Repartitionin
g cluster data, Migration tasks count: 108
hz-node-2-1 | 2026-02-25 12:19:57,083 [ INFO] [hz.focused_tharp.migration] [c.h.i.p.i.MigrationManagerImpl]: [172.18.0.4]:5701 [dev] [5.6.0] All migration
tasks have been completed. (repartitionTime=Wed Feb 25 12:19:56 GMT 2026, plannedMigrations=108, completedMigrations=108, remainingMigrations=0, totalComp
letedMigrations=2088)
hz-node-1-1 | 2026-02-25 12:21:23,949 [ INFO] [hz.focused_tharp.cached.thread-5] [c.h.i.s.t.TcpServerConnector]: [172.18.0.4]:5701 [dev] [5.6.0] Connectin
g to /172.18.0.5:5701, timeout: 10000, bind-any: true
hz-node-2-1 | 2026-02-25 12:21:33,951 [ INFO] [hz.focused_tharp.cached.thread-5] [c.h.i.s.t.TcpServerConnector]: [172.18.0.4]:5701 [dev] [5.6.0] Could not
connect to: /172.18.0.5:5701, Reason: IOException[Connect timed out to address /172.18.0.5:5701]
hz-node-2-1 | 2026-02-25 12:21:34,054 [ INFO] [hz.focused_tharp.cached.thread-4] [c.h.i.s.t.TcpServerConnector]: [172.18.0.4]:5701 [dev] [5.6.0] Connectin
g to /172.18.0.5:5701, timeout: 10000, bind-any: true
hz-node-2-1 | 2026-02-25 12:21:44,068 [ INFO] [hz.focused_tharp.cached.thread-4] [c.h.i.s.t.TcpServerConnector]: [172.18.0.4]:5701 [dev] [5.6.0] Could not
connect to: /172.18.0.5:5701, Reason: IOException[Connect timed out to address /172.18.0.5:5701]
hz-node-2-1 | 2026-02-25 12:21:44,069 [ WARN] [hz.focused_tharp.cached.thread-4] [c.h.i.s.t.TcpServerConnectionErrorHandler]: [172.18.0.4]:5701 [dev] [5.6
.0] Removing connection to endpoint [172.18.0.5]:5701 Cause => java.io.IOException[Connect timed out to address /172.18.0.5:5701], Error-Count: 5
hz-node-2-1 | 2026-02-25 12:21:44,070 [ INFO] [hz.focused_tharp.cached.thread-4] [c.h.i.c.i.MembershipManager]: [172.18.0.4]:5701 [dev] [5.6.0] Removing M
ember [172.18.0.5]:5701 - 66aa9bd9-f8ee-472b-b5df-fd61f6a2d396
hz-node-2-1 | 2026-02-25 12:21:44,070 [ INFO] [hz.focused_tharp.cached.thread-4] [c.h.i.p.i.PartitionStateManagerImpl]: [172.18.0.4]:5701 [dev] [5.6.0] St
oring snapshot of partition assignments while removing UUID 66aa9bd9-f8ee-472b-b5df-fd61f6a2d396
hz-node-2-1 | 2026-02-25 12:21:44,073 [ INFO] [hz.focused_tharp.cached.thread-4] [c.h.i.c.ClusterService]: [172.18.0.4]:5701 [dev] [5.6.0]
hz-node-2-1 | Members (size:1, ver:21) [
hz-node-2-1 | Member [172.18.0.4]:5701 - 5e74a703-9d11-4f7e-8436-7cb783a7119c this
hz-node-2-1 | ]
hz-node-2-1 | 2026-02-25 12:21:44,074 [ INFO] [hz.focused_tharp.cached.thread-10] [c.h.t.TransactionManagerService]: [172.18.0.4]:5701 [dev] [5.6.0] Commi
tting/rolling-back live transactions of [172.18.0.5]:5701, UUID: 66aa9bd9-f8ee-472b-b5df-fd61f6a2d396
hz-node-1-1 | 2026-02-25 12:21:44,077 [ INFO] [MC-Client-dev.event-3] [c.h.w.s.MCClientManager]: Member removed: Member [172.18.0.5]:5701 - 66aa9bd9-f8ee-
472b-b5df-fd61f6a2d396
hz-node-2-1 | 2026-02-25 12:21:44,088 [ INFO] [hz.focused_tharp.migration] [c.h.i.p.i.MigrationManagerImpl]: [172.18.0.4]:5701 [dev] [5.6.0] Successfully
promoted 141 backups on 1 members in 11 ms
hz-node-2-1 | 2026-02-25 12:21:44,090 [ INFO] [hz.focused_tharp.migration] [c.h.i.p.i.MigrationManagerImpl]: [172.18.0.4]:5701 [dev] [5.6.0] Partition bal
ance is ok, no need to repartition.
```

Розподілення даних по робочих нодах зайняло деякий час, але тут втрата даних відсутня.

3. Тут я зупинила спочатку одну ноду, після розподілу зробила те саме для іншої. Дані перерозподілились на єдину робочу ноду і втрат немає

```
(facade-service) ira@MacBook-Air-Ira software_architecture % docker-compose stop hz-node-3
[+] stop 1/1
✓ Container software_architecture-hz-node-3-1 Stopped
(facade-service) ira@MacBook-Air-Ira software_architecture % docker-compose stop hz-node-1
[+] stop 1/1
✓ Container software_architecture-hz-node-1-1 Stopped
```

Member	Entries	Gets	Puts	Removals	Sets	Ents
172.18.0.4:5701	1,000	0	2,564	0	0	
TOTAL	1,000	0	2,888	0	0	

```
hz-node-1-1 exited with code 143
hz-node-2-1 | 2026-02-25 12:21:23,949 [ INFO] [hz.focused_tharp.cached.thread-5] [c.h.i.s.t.TcpServerConnector]: [172.18.0.4]:5701 [dev] [5.6.0] Connectin
g to /172.18.0.5:5701, timeout: 10000, bind-any: true
hz-node-2-1 | 2026-02-25 12:21:33,951 [ INFO] [hz.focused_tharp.cached.thread-5] [c.h.i.s.t.TcpServerConnector]: [172.18.0.4]:5701 [dev] [5.6.0] Could not
connect to: /172.18.0.5:5701, Reason: IOException[Connect timed out to address /172.18.0.5:5701]
hz-node-2-1 | 2026-02-25 12:21:34,054 [ INFO] [hz.focused_tharp.cached.thread-4] [c.h.i.s.t.TcpServerConnector]: [172.18.0.4]:5701 [dev] [5.6.0] Connectin
g to /172.18.0.5:5701, timeout: 10000, bind-any: true
hz-node-2-1 | 2026-02-25 12:21:44,068 [ INFO] [hz.focused_tharp.cached.thread-4] [c.h.i.s.t.TcpServerConnector]: [172.18.0.4]:5701 [dev] [5.6.0] Could not
connect to: /172.18.0.5:5701, Reason: IOException[Connect timed out to address /172.18.0.5:5701]
hz-node-2-1 | 2026-02-25 12:21:44,069 [ WARN] [hz.focused_tharp.cached.thread-4] [c.h.i.s.t.TcpServerConnectionErrorHandler]: [172.18.0.4]:5701 [dev] [5.6
.0] Removing connection to endpoint [172.18.0.5]:5701 Cause => java.io.IOException[Connect timed out to address /172.18.0.5:5701], Error-Count: 5
hz-node-2-1 | 2026-02-25 12:21:44,070 [ INFO] [hz.focused_tharp.cached.thread-4] [c.h.i.c.i.MembershipManager]: [172.18.0.4]:5701 [dev] [5.6.0] Removing M
ember [172.18.0.5]:5701 - 66aa9bd9-f8ee-472b-b5df-fd61f6a2d396
hz-node-2-1 | 2026-02-25 12:21:44,070 [ INFO] [hz.focused_tharp.cached.thread-4] [c.h.i.p.i.PartitionStateManagerImpl]: [172.18.0.4]:5701 [dev] [5.6.0] St
oring snapshot of partition assignments while removing UUID 66aa9bd9-f8ee-472b-b5df-fd61f6a2d396
hz-node-2-1 | 2026-02-25 12:21:44,073 [ INFO] [hz.focused_tharp.cached.thread-4] [c.h.i.c.ClusterService]: [172.18.0.4]:5701 [dev] [5.6.0]
hz-node-2-1 | Members (size:1, ver:21) [
hz-node-2-1 | Member [172.18.0.4]:5701 - 5e74a703-9d11-4f7e-8436-7cb783a7119c this
hz-node-2-1 | ]
hz-node-2-1 | 2026-02-25 12:21:44,074 [ INFO] [hz.focused_tharp.cached.thread-10] [c.h.t.TransactionManagerService]: [172.18.0.4]:5701 [dev] [5.6.0] Commi
tting/rolling-back live transactions of [172.18.0.5]:5701, UUID: 66aa9bd9-f8ee-472b-b5df-fd61f6a2d396
hz-node-1-1 | 2026-02-25 12:21:44,077 [ INFO] [MC-Client-dev.event-3] [c.h.w.s.MCClientManager]: Member removed: Member [172.18.0.5]:5701 - 66aa9bd9-f8ee-
472b-b5df-fd61f6a2d396
hz-node-2-1 | 2026-02-25 12:21:44,088 [ INFO] [hz.focused_tharp.migration] [c.h.i.p.i.MigrationManagerImpl]: [172.18.0.4]:5701 [dev] [5.6.0] Successfully
promoted 141 backups on 1 members in 11 ms
hz-node-2-1 | 2026-02-25 12:21:44,090 [ INFO] [hz.focused_tharp.migration] [c.h.i.p.i.MigrationManagerImpl]: [172.18.0.4]:5701 [dev] [5.6.0] Partition bal
ance is ok, no need to repartition.
```

4. Наступним кроком є відключити одночасно 2 ноди, емулюючи падіння серверів.
`docker-compose kill hz-node-1 hz-node-3`

```
hz-node-2-1 | 2026-02-25 12:04:50,617 [ INFO] [hz.focused_tharp.cached.thread-7] [c.h.t.TransactionManagerService]: [172.18.0.4]:5701 [dev] [5.6.0] Committing/rolling-back live transactions of [172.18.0.5]:5701, UUID: 753cac22-4aac-4a5d-be79-acffe3735811
mancenter-1 | 2026-02-25 12:04:50,620 [ INFO] [MC-Client-dev.event-1] [c.h.w.s.MCClientManager]: Member removed: Member [172.18.0.5]:5701 - 753cac22-4aac-4a5d-be79-acffe3735811
hz-node-2-1 | 2026-02-25 12:04:57,575 [ INFO] [hz.focused_tharp.cached.thread-11] [c.h.i.p.InternalPartitionService]: [172.18.0.4]:5701 [dev] [5.6.0] Remaining migration tasks: 1. (repartitionTime=Wed Feb 25 12:00:37 GMT 2026, plannedMigrations=271, completedMigrations=271, remainingMigrations=0, totalCompletedMigrations=2164)
hz-node-2-1 | 2026-02-25 12:05:00,548 [ INFO] [hz.focused_tharp.cached.thread-3] [c.h.i.s.t.TcpServerConnector]: [172.18.0.4]:5701 [dev] [5.6.0] Connecting to /172.18.0.7:5701, timeout: 10000, bind-any: true
hz-node-2-1 | 2026-02-25 12:05:03,664 [ INFO] [hz.focused_tharp.cached.thread-3] [c.h.i.s.t.TcpServerConnector]: [172.18.0.4]:5701 [dev] [5.6.0] Could not connect to: /172.18.0.7:5701. Reason: IOException[Host is unreachable to address /172.18.0.7:5701]
hz-node-2-1 | 2026-02-25 12:05:06,970 [ INFO] [hz.focused_tharp.cached.thread-12] [c.h.i.s.t.TcpServerConnector]: [172.18.0.4]:5701 [dev] [5.6.0] Connecting to /172.18.0.7:5701, timeout: 10000, bind-any: true
hz-node-2-1 | 2026-02-25 12:05:10,066 [ INFO] [hz.focused_tharp.cached.thread-12] [c.h.i.s.t.TcpServerConnector]: [172.18.0.4]:5701 [dev] [5.6.0] Could not connect to: /172.18.0.7:5701. Reason: IOException[Host is unreachable to address /172.18.0.7:5701]
hz-node-2-1 | 2026-02-25 12:05:10,067 [ WARN] [hz.focused_tharp.cached.thread-12] [c.h.i.s.t.TcpServerConnectionErrorHandler]: [172.18.0.4]:5701 [dev] [5.6.0] Removing connection to endpoint [172.18.0.7]:5701 Cause => java.io.IOException [Host is unreachable to address /172.18.0.7:5701], Error-Count: 5
hz-node-2-1 | 2026-02-25 12:05:10,067 [ INFO] [hz.focused_tharp.cached.thread-12] [c.h.i.c.i.MembershipManager]: [172.18.0.4]:5701 [dev] [5.6.0] Removing Member [172.18.0.7]:5701 - efaa3242-7742-4d5a-b335-6f4bcd7c9aa6
hz-node-2-1 | 2026-02-25 12:05:10,067 [ INFO] [hz.focused_tharp.cached.thread-12] [c.h.i.p.i.PartitionStateManagerImpl]: [172.18.0.4]:5701 [dev] [5.6.0] Storing snapshot of partition assignments while removing UUID efaa3242-7742-4d5a-b335-6f4bcd7c9aa6
hz-node-2-1 | 2026-02-25 12:05:10,068 [ INFO] [hz.focused_tharp.cached.thread-12] [c.h.i.c.ClusterService]: [172.18.0.4]:5701 [dev] [5.6.0]
hz-node-2-1 | Members {size:1, ver:17} [
hz-node-2-1 |   Member [172.18.0.4]:5701 - 5e74a703-9d11-4f7e-8436-7cb783a7119c this
hz-node-2-1 | ]
hz-node-2-1 | 2026-02-25 12:05:10,069 [ INFO] [hz.focused_tharp.cached.thread-9] [c.h.t.TransactionManagerService]: [172.18.0.4]:5701 [dev] [5.6.0] Committing/rolling-back live transactions of [172.18.0.7]:5701, UUID: efaa3242-7742-4d5a-b335-6f4bcd7c9aa6
mancenter-1 | 2026-02-25 12:05:10,069 [ INFO] [MC-Client-dev.event-2] [c.h.w.s.MCClientManager]: Member removed: Member [172.18.0.7]:5701 - efaa3242-7742-4d5a-b335-6f4bcd7c9aa6
hz-node-2-1 | 2026-02-25 12:05:10,074 [ WARN] [hz.focused_tharp.migration] [c.h.i.p.i.MigrationManagerImpl]: [172.18.0.4]:5701 [dev] [5.6.0] Promotion commit failed for 44 migrations since destination [172.18.0.7]:5701 - efaa3242-7742-4d5a-b335-6f4bcd7c9aa6 left the cluster
hz-node-2-1 | 2026-02-25 12:05:10,075 [ INFO] [hz.focused_tharp.migration] [c.h.i.p.i.MigrationManagerImpl]: [172.18.0.4]:5701 [dev] [5.6.0] Tried to promote 90 backups on 2 members but some failed in 19455 ms
hz-node-2-1 | 2026-02-25 12:05:10,084 [ INFO] [hz.focused_tharp.migration] [c.h.i.p.i.MigrationManagerImpl]: [172.18.0.4]:5701 [dev] [5.6.0] Successfully promoted 134 backups on 1 members in 8 ms
hz-node-2-1 | 2026-02-25 12:05:10,086 [ INFO] [hz.focused_tharp.migration] [c.h.i.p.i.MigrationManagerImpl]: [172.18.0.4]:5701 [dev] [5.6.0] Partition balance is ok, no need to repartition.
```

З логів вище можна побачити, що кластер помітив, що нода з IP 172.18.0.5 зникла. Жива нода успішно завершила 271 завдання з міграції (`completedMigrations=271`). Це означає, що вона встигла забрати собі копії даних, які належали першій втраченій ноді. На цей момент кластер ще вважав, що друга нода жива. Далі нода `hz-node-2` намагається зв'язатися з 172.18.0.7, але отримує помилку: `Host is unreachable`. Видаляє останнього учасника зі списку: `Removing Member [172.18.0.7]`. Далі є такий рядок `Promotion commit failed for 44 migrations since destination ... left the cluster`. Коли була вбита друга нода, вона не встигла передати 44 фрагменти даних. Жива нода намагалася їх підхопити, але оскільки цільовий об'єкт зник посеред процесу, ці 44 партиції просто випали з пам'яті.

5. Тому так, тут є втрата даних.

Member	Entries	Gets	Puts	Removals	Sets	Entries
172.18.0.4:5701	671	0	2,907	0	0	
TOTAL	671	0	2,907	0	0	

6. Що зробити, щоб не було втрати даних? Щоб дані вижили навіть при одночасному падінні 2-х нод, має бути 2 копії (бекапи) кожного запису. Тоді дані існують на всіх трьох нодах одночасно (1 оригінал + 2 бекапи).

```
hz-node-1:
image: hazelcast/hazelcast:5.6
ports: ["5701:5701"]
volumes:
- ./hazelcast.yaml:/opt/hazelcast/config/hazelcast.yaml
environment:
- JAVA_OPTS=-Dhazelcast.config=/opt/hazelcast/config/hazelcast.yaml
```

```
hazelcast:
cluster-name: dev
network:
join:
tcp-ip:
enabled: true
member-list:
- hz-node-1
- hz-node-2
- hz-node-3
map:
testmap:
backup-count: 2
async-backup-count: 0
```

Для того я дещо переписала `docker-compose` та написала файл

hazelcast.yml. Саме в ньому я для кожної ноди вказала 2 бекапи.
І docker-compose тепер використовує його як конфіг(-Dhazelcast.config).
Тепер якщо зайти у management center -> testmap -> map config можна справді побачити
що 2 бекапи створюються.

Map Config

Backup Count:	Async Backup Count:	In-Memory Storage Format:
2	0	BINARY
Max Size:	Max Size Policy:	Max Idle (seconds):
2147483647	PER_NODE	0
TTL (seconds):	Eviction Policy:	☐ Read Backup Data
0	NONE	
Merge Policy:		
com.hazelcast.spi.merge.PutIfAbsentMergePolicy		

Далі я спробувала знов відключити одночасно 2 ноди:
docker-compose kill hz-node-1 hz-node-3

Member	Entries	Gets	Puts	Removals	Sets	Enti
hz-node-1:5701	336	0	336	0	0	
hz-node-2:5701	1,000	0	683	0	0	
hz-node-3:5701	324	0	324	0	0	
TOTAL	1,660	0	1,343	0	0	

Можна побачити, що під час перерозподіленні даних, total значення ставало більше 1000.

```
hz-node-2-1 | 2026-02-25 14:44:04,686 [ INFO] [hz.stupefied_mclaren.cached.thread-1] [c.h.i.c.i.MembershipManager]: [
hz-node-2]:5701 [dev] [5.6.0] Mastership is claimed with: MembersView{version=4, members=[MemberInfo{address=[hz-node-2]:570
1, uuid=cabb9b24-13b6-465d-b156-0f95f58ee3ed, cpMemberUUID=null, liteMember=false, memberListJoinVersion=2}]}
mancenter-1 | 2026-02-25 14:44:04,686 [ INFO] [MC-Client-dev.event-1] [c.h.w.s.MCClientManager]: Member removed: Memb
er [hz-node-1]:5701 - 7025c789-e561-45f2-9d3e-8b827cd633af
hz-node-2-1 | 2026-02-25 14:44:04,689 [ INFO] [hz.stupefied_mclaren.cached.thread-6] [c.h.t.TransactionManagerService
]: [hz-node-2]:5701 [dev] [5.6.0] Committing/rolling-back live transactions of [hz-node-1]:5701, UUID: 7025c789-e561-45f2-9d
3e-8b827cd633af
hz-node-2-1 | 2026-02-25 14:44:04,692 [ INFO] [hz.stupefied_mclaren.cached.thread-6] [c.h.t.TransactionManagerService
]: [hz-node-2]:5701 [dev] [5.6.0] Committing/rolling-back live transactions of [hz-node-3]:5701, UUID: d019af65-6313-4e61-9d
46-b1f46c33a99c
mancenter-1 | 2026-02-25 14:44:04,704 [ INFO] [MC-Client-dev.event-1] [c.h.w.s.MCClientManager]: Member removed: Memb
er [hz-node-3]:5701 - d019af65-6313-4e61-9d46-b1f46c33a99c
hz-node-2-1 | 2026-02-25 14:44:04,938 [ INFO] [hz.stupefied_mclaren.migration] [c.h.i.p.InternalPartitionService]: [h
z-node-2]:5701 [dev] [5.6.0] Fetching partition tables from cluster to determine the most recent one... Local stamp: 8082085
890800019451
hz-node-2-1 | 2026-02-25 14:44:04,939 [ INFO] [hz.stupefied_mclaren.migration] [c.h.i.p.InternalPartitionService]: [h
z-node-2]:5701 [dev] [5.6.0] Most recent partition table is determined.
hz-node-2-1 | 2026-02-25 14:44:04,939 [ INFO] [hz.stupefied_mclaren.migration] [c.h.i.p.InternalPartitionService]: [h
z-node-2]:5701 [dev] [5.6.0] Applying the most recent of partition state...
hz-node-2-1 | 2026-02-25 14:44:04,943 [ WARN] [hz.stupefied_mclaren.migration] [c.h.i.p.InternalPartitionService]: [h
z-node-2]:5701 [dev] [5.6.0] Following unknown addresses are found in partition table sent from master[[hz-node-2]:5701]. (P
robably they have recently joined or left the cluster.) {
hz-node-2-1 | [hz-node-1]:5701 - 7025c789-e561-45f2-9d3e-8b827cd633af
hz-node-2-1 | [hz-node-3]:5701 - d019af65-6313-4e61-9d46-b1f46c33a99c
hz-node-2-1 | }
hz-node-2-1 | 2026-02-25 14:44:05,021 [ INFO] [hz.stupefied_mclaren.migration] [c.h.i.p.i.MigrationManagerImpl]: [hz-
node-2]:5701 [dev] [5.6.0] Successfully promoted 180 backups on 1 members in 75 ms
hz-node-2-1 | 2026-02-25 14:44:05,024 [ INFO] [hz.stupefied_mclaren.migration] [c.h.i.p.i.MigrationManagerImpl]: [hz-
node-2]:5701 [dev] [5.6.0] Partition balance is ok, no need to repartition.
```


І ось результат, де всі дані збережено.

Member ^	Entries	Gets	Puts	Removals	Sets	Enti
hz-node-2:5701	1,000	0	683	0	0	
TOTAL	1,000	0	683	0	0	

Крок 4. Продемонструвати роботу Distributed Map without locks

Тут я написала код що використовує Distributed map без блокувань. Клієнти запускаються як 3 паралельних процеси і виконують роботу написану у функції `worker()`. Кожен такий клієнт створює власний Hazelcast client і підключається до кластера з 3 нод. Кожен інкрементує одне й те саме значення 10 000 разів і очікуваним результатом є 30 000. В циклі клієнт читає значення з кластера, збільшує його і записує назад. Такі операції не атомарні, тому виникатиме race condition та дані не будуть консистентними. Для запуску:

```
docker-compose exec counter-service uv run python task4.py
```

```
Client 2 started
Client 1 started
Client 0 started
Client 2 finished
Client 1 finished
Client 0 finished
FINAL VALUE: 13375
```

Кінцеве значення $13375 < 30000$, хоча при кожному запуску воно може бути іншим. Тобто тут race condition і справді є проблемою, бо дані втрачаються.

```
hz-node-2-1 | 2026-02-25 20:28:50,271 [ INFO] [hz.priceless_sanderson.event-3] [c.h.c.i.ClientEndpointManager]: [hz-node-2]:5701 [dev] [5.6.0] Destroying ClientEndpoint{connection=Connection[id=37, /172.18.0.6:5701->software_architecture-counter-service-1.software_architecture_default/172.18.0.3:34126, qualifier=null, endpoint=[172.18.0.3]:34126, remoteUid=75eea9fa-5c7f-4425-b20d-8570ba8aa7a9, alive=false, connectionType=PYH, planeIndex=-1], clientUid=75eea9fa-5c7f-4425-b20d-8570ba8aa7a9, clientName=hz.client_0, authenticated=true, clientVersion=5.6.0, creationTime=1772051324933, latest clientAttributes=null, labels=[]}
hz-node-3-1 | 2026-02-25 20:28:50,272 [ INFO] [hz.serene_euclid.event-5] [c.h.c.i.ClientEndpointManager]: [hz-node-3]:5701 [dev] [5.6.0] Destroying ClientEndpoint{connection=Connection[id=39, /172.18.0.5:5701->software_architecture-counter-service-1.software_architecture_default/172.18.0.3:37614, qualifier=null, endpoint=[172.18.0.3]:37614, remoteUid=75eea9fa-5c7f-4425-b20d-8570ba8aa7a9, alive=false, connectionType=PYH, planeIndex=-1], clientUid=75eea9fa-5c7f-4425-b20d-8570ba8aa7a9, clientName=hz.client_0, authenticated=true, clientVersion=5.6.0, creationTime=1772051324943, latest clientAttributes=null, labels=[]}
hz-node-2-1 | 2026-02-25 20:28:50,275 [ INFO] [hz.priceless_sanderson.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [hz-node-2]:5701 [dev] [5.6.0] Received auth from Connection[id=40, /172.18.0.6:5701->/172.18.0.3:34168, qualifier=null, endpoint=[172.18.0.3]:34168, remoteUid=338dc862-4d17-4a8c-aa8d-866beb886b74, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: 338dc862-4d17-4a8c-aa8d-866beb886b74, client name: hz.client_0, client version: 5.6.0
hz-node-1-1 | 2026-02-25 20:28:50,288 [ INFO] [hz.zealous_galois.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [hz-node-1]:5701 [dev] [5.6.0] Received auth from Connection[id=56, /172.18.0.2:5701->/172.18.0.3:44496, qualifier=null, endpoint=[172.18.0.3]:44496, remoteUid=338dc862-4d17-4a8c-aa8d-866beb886b74, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: 338dc862-4d17-4a8c-aa8d-866beb886b74, client name: hz.client_0, client version: 5.6.0
hz-node-3-1 | 2026-02-25 20:28:50,305 [ INFO] [hz.serene_euclid.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [hz-node-3]:5701 [dev] [5.6.0] Received auth from Connection[id=40, /172.18.0.5:5701->/172.18.0.3:37622, qualifier=null, endpoint=[172.18.0.3]:37622, remoteUid=338dc862-4d17-4a8c-aa8d-866beb886b74, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: 338dc862-4d17-4a8c-aa8d-866beb886b74, client name: hz.client_0, client version: 5.6.0
hz-node-3-1 | 2026-02-25 20:28:50,322 [ INFO] [hz.serene_euclid.IO.thread-in-0] [c.h.i.s.t.TcpServerConnection]: [hz-node-3]:5701 [dev] [5.6.0] Connection[id=40, /172.18.0.5:5701->/172.18.0.3:37622, qualifier=null, endpoint=[172.18.0.3]:37622, remoteUid=338dc862-4d17-4a8c-aa8d-866beb886b74, alive=false, connectionType=PYH, planeIndex=-1] closed. Reason: Connection closed by the other side
hz-node-1-1 | 2026-02-25 20:28:50,322 [ INFO] [hz.zealous_galois.IO.thread-in-1] [c.h.i.s.t.TcpServerConnection]: [hz-node-1]:5701 [dev] [5.6.0] Connection[id=56, /172.18.0.2:5701->/172.18.0.3:44496, qualifier=null, endpoint=[172.18.0.3]:44496, remoteUid=338dc862-4d17-4a8c-aa8d-866beb886b74, alive=false, connectionType=PYH, planeIndex=-1] closed. Reason: Connection closed by the other side
hz-node-2-1 | 2026-02-25 20:28:50,322 [ INFO] [hz.priceless_sanderson.IO.thread-in-0] [c.h.i.s.t.TcpServerConnection]: [hz-node-2]:5701 [dev] [5.6.0] Connection[id=40, /172.18.0.6:5701->/172.18.0.3:34168, qualifier=null, endpoint=[172.18.0.3]:34168, remoteUid=338dc862-4d17-4a8c-aa8d-866beb886b74, alive=false, connectionType=PYH, planeIndex=-1] closed. Reason: Connection closed by the other side
```

Крок 5. Робота Distributed Map with pessimistic locking

Тут вся логіка збережена з кроку 4, але тепер я використовую блокування. Доступ до ресурсу блокується процесом для інших доки він не завершить роботу.

```
docker-compose exec counter-service uv run python task5.py
```

```
Client 1 started
Client 0 started
Client 2 started
Client 0 finished in 22.61 seconds
Client 1 finished in 22.61 seconds
Client 2 finished in 22.61 seconds
FINAL VALUE: 30000
TOTAL EXECUTION TIME: 22.65 seconds
```

Результат тепер і справді як і очікувалось 30000, загальний час для песимістичного блокування 22.65 секунд.

```
hz-node-1-1 | 2026-02-25 20:34:07,038 [ INFO] [hz.zealous_galois.IO.thread-in-0] [c.h.i.s.t.TcpServerConnection]: [hz-node-1]:5701 [dev] [5.6.0] Connection[id=64, /172.18.0.2:5701->/172.18.0.3:42484, qualifier=null, endpoint=[172.18.0.3]:42484, remoteUid=d096a380-b30f-4455-894c-bf5ec504f933, alive=false, connectionType=PYH, planeIndex=-1] closed. Reason: Connection closed by the other side
hz-node-2-1 | 2026-02-25 20:34:07,039 [ INFO] [hz.priceless_sanderson.IO.thread-in-2] [c.h.i.s.t.TcpServerConnection]: [hz-node-2]:5701 [dev] [5.6.0] Connection[id=48, /172.18.0.6:5701->/172.18.0.3:54172, qualifier=null, endpoint=[172.18.0.3]:54172, remoteUid=d096a380-b30f-4455-894c-bf5ec504f933, alive=false, connectionType=PYH, planeIndex=-1] closed. Reason: Connection closed by the other side
hz-node-1-1 | 2026-02-25 20:34:07,039 [ INFO] [hz.zealous_galois.event-5] [c.h.c.i.ClientEndpointManager]: [hz-node-1]:5701 [dev] [5.6.0] Destroying ClientEndpoint(connection=Connection[id=64, /172.18.0.2:5701->/172.18.0.3:42484, qualifier=null, endpoint=[172.18.0.3]:42484, remoteUid=d096a380-b30f-4455-894c-bf5ec504f933, alive=false, connectionType=PYH, planeIndex=-1], clientUid=d096a380-b30f-4455-894c-bf5ec504f933, clientName=hz.client_0, authenticated=true, clientVersion=5.6.0, creationTime=1772051647024, latest clientAttributes=null, labels={})
hz-node-2-1 | 2026-02-25 20:34:07,039 [ INFO] [hz.priceless_sanderson.event-2] [c.h.c.i.ClientEndpointManager]: [hz-node-2]:5701 [dev] [5.6.0] Destroying ClientEndpoint(connection=Connection[id=48, /172.18.0.6:5701->/172.18.0.3:54172, qualifier=null, endpoint=[172.18.0.3]:54172, remoteUid=d096a380-b30f-4455-894c-bf5ec504f933, alive=false, connectionType=PYH, planeIndex=-1], clientUid=d096a380-b30f-4455-894c-bf5ec504f933, clientName=hz.client_0, authenticated=true, clientVersion=5.6.0, creationTime=1772051647025, latest clientAttributes=null, labels={})
hz-node-3-1 | 2026-02-25 20:34:10,908 [ WARN] [hz.serene_euclid.cached.thread-7] [c.h.c.i.ClientEngine]: [hz-node-3]:5701 [dev] [5.6.0] No connection left to client cluster wide 2d18f326-f529-421e-ab0a-96ddede12918 for 3899 millis, cleaning resources of the client
hz-node-3-1 | 2026-02-25 20:34:10,908 [ INFO] [hz.serene_euclid.cached.thread-7] [c.h.t.TransactionManagerService]: [hz-node-3]:5701 [dev] [5.6.0] Committing/rolling-back live transactions of client, UUID: 2d18f326-f529-421e-ab0a-96ddede12918
hz-node-3-1 | 2026-02-25 20:34:10,909 [ WARN] [hz.serene_euclid.cached.thread-7] [c.h.c.i.ClientEngine]: [hz-node-3]:5701 [dev] [5.6.0] No connection left to client cluster wide b2902a52-73c6-4b26-90b0-7700fc0f71a4 for 3904 millis, cleaning resources of the client
hz-node-3-1 | 2026-02-25 20:34:10,909 [ INFO] [hz.serene_euclid.cached.thread-7] [c.h.t.TransactionManagerService]: [hz-node-3]:5701 [dev] [5.6.0] Committing/rolling-back live transactions of client, UUID: b2902a52-73c6-4b26-90b0-7700fc0f71a4
hz-node-3-1 | 2026-02-25 20:34:10,909 [ WARN] [hz.serene_euclid.cached.thread-7] [c.h.c.i.ClientEngine]: [hz-node-3]:5701 [dev] [5.6.0] No connection left to client cluster wide cacfe8ea-743d-47f0-ab0a-34b2e469b0d4 for 3901 millis, cleaning resources of the client
hz-node-3-1 | 2026-02-25 20:34:10,910 [ INFO] [hz.serene_euclid.cached.thread-7] [c.h.t.TransactionManagerService]: [hz-node-3]:5701 [dev] [5.6.0] Committing/rolling-back live transactions of client, UUID: cacfe8ea-743d-47f0-ab0a-34b2e469b0d4
hz-node-3-1 | 2026-02-25 20:34:07,020 [ INFO] [hz.serene_euclid.IO.thread-in-1] [c.h.i.s.t.TcpServerConnection]: [hz-node-3]:5701 [dev] [5.6.0] Connection[id=47, /172.18.0.5:5701->software_architecture-counter-service-1.software_architecture_default/172.18.0.3:32938, qualifier=null, endpoint=[172.18.0.3]:32938, remoteUid=cacfe8ea-743d-47f0-ab0a-34b2e469b0d4, alive=false, connectionType=PYH, planeIndex=-1] closed. Reason: Connection closed by the other side
hz-node-2-1 | 2026-02-25 20:34:07,019 [ INFO] [hz.priceless_sanderson.event-5] [c.h.c.i.ClientEndpointManager]: [hz-node-2]:5701 [dev] [5.6.0] Destroying ClientEndpoint(connection=Connection[id=45, /172.18.0.6:5701->software_architecture-counter-service-1.software_architecture_default/172.18.0.3:51054, qualifier=null, endpoint=[172.18.0.3]:51054, remoteUid=2d18f326-f529-421e-ab0a-96ddede12918, alive=false, connectionType=PYH, planeIndex=-1], clientUid=2d18f326-f529-421e-ab0a-96ddede12918, clientName=hz.client_0, authenticated=true, clientVersion=5.6.0, creationTime=1772051624746, latest clientAttributes=null, labels={})
hz-node-1-1 | 2026-02-25 20:34:07,018 [ INFO] [hz.zealous_galois.event-1] [c.h.c.i.ClientEndpointManager]: [hz-node-1]:5701 [dev] [5.6.0] Destroying ClientEndpoint(connection=Connection[id=63, /172.18.0.2:5701->software_architecture-counter-service-1.software_architecture_default/172.18.0.3:43678, qualifier=null, endpoint=[172.18.0.3]:43678, remoteUid=b2902a52-73c6-4b26-90b0-7700fc0f71a4, alive=false, connectionType=PYH, planeIndex=-1], clientUid=b2902a52-73c6-4b26-90b0-7700fc0f71a4, clientName=hz.client_0, authenticated=true, clientVersion=5.6.0, creationTime=1772051624752, latest clientAttributes=null, labels={})
hz-node-3-1 | 2026-02-25 20:34:07,021 [ INFO] [hz.serene_euclid.event-1] [c.h.c.i.ClientEndpointManager]: [hz-node-3]:5701 [dev] [5.6.0] Destroying ClientEndpoint(connection=Connection[id=47, /172.18.0.5:5701->software_architecture-counter-service-1.software_architecture_default/172.18.0.3:32938, qualifier=null, endpoint=[172.18.0.3]:32938, remoteUid=cacfe8ea-743d-47f0-ab0a-34b2e469b0d4, alive=false, connectionType=PYH, planeIndex=-1], clientUid=cacfe8ea-743d-47f0-ab0a-34b2e469b0d4, clientName=hz.client_0, authenticated=true, clientVersion=5.6.0, creationTime=1772051624755, latest clientAttributes=null, labels={})
hz-node-1-1 | 2026-02-25 20:34:07,020 [ INFO] [hz.zealous_galois.IO.thread-in-0] [c.h.i.s.t.TcpServerConnection]: [hz-node-1]:5701 [dev] [5.6.0] Connection[id=61, /172.18.0.2:5701->software_architecture-counter-service-1.software_architecture_default/172.18.0.3:43654, qualifier=null, endpoint=[172.18.0.3]:43654, remoteUid=cacfe8ea-743d-47f0-ab0a-34b2e469b0d4, alive=false, connectionType=PYH, planeIndex=-1] closed. Reason: Connection closed by the other side
hz-node-2-1 | 2026-02-25 20:34:07,020 [ INFO] [hz.priceless_sanderson.IO.thread-in-2] [c.h.i.s.t.TcpServerConnection]: [hz-node-2]:5701 [dev] [5.6.0] Connection[id=46, /172.18.0.6:5701->software_architecture-counter-service-1.software_architecture_default/172.18.0.3:51066, qualifier=null, endpoint=[172.18.0.3]:51066, remoteUid=cacfe8ea-743d-47f0-ab0a-34b2e469b0d4, alive=false, connectionType=PYH, planeIndex=-1] closed. Reason: Connection closed by the other side
hz-node-1-1 | 2026-02-25 20:34:07,020 [ INFO] [hz.zealous_galois.event-5] [c.h.c.i.ClientEndpointManager]: [hz-node-1]:5701 [dev] [5.6.0] Destroying ClientEndpoint(connection=Connection[id=61, /172.18.0.2:5701->software_architecture-counter-service-1.software_architecture_default/172.18.0.3:43654, qualifier=null, endpoint=[172.18.0.3]:43654, remoteUid=cacfe8ea-743d-47f0-ab0a-34b2e469b0d4, alive=false, connectionType=PYH, planeIndex=-1], clientUid=cacfe8ea-743d-47f0-ab0a-34b2e469b0d4, clientName=hz.client_0, authenticated=true, clientVersion=5.6.0, creationTime=1772051624747, latest clientAttributes=null, labels={})
hz-node-3-1 | 2026-02-25 20:34:07,023 [ INFO] [hz.serene_euclid.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticatingMessageTask]: [hz-node-3]:5701 [dev] [5.6.0] Received auth from Connection[id=48, /172.18.0.5:5701->/172.18.0.3:51674, qualifier=null, endpoint=[172.18.0.3]:51674, remoteUid=d096a380-b30f-4455-894c-bf5ec504f933, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: d096a380-b30f-4455-894c-bf5ec504f933, client name: hz.client_0, client version: 5.6.0
```

Крок 6. Робота Distributed Map with optimistic locking

Знову ж таки, логіка роботи не змінюється, але ідея оптимістичного блокування у тому, що ймовірно конфлікти будуть траплятися рідко, тому замість блокування ресурсу при конфлікті просто треба повторити операцію. Ніхто нікого не блокує, всі можуть читати й намагатися писати, а система перевіряє, чи значення не змінилось. Якщо змінилось - то retry. Реалізовано це через атомарну операцію `replace_if_same(key, old_value, new_value)`, якщо поточне значення `== old_value` -> замінити, якщо ні, то нічого не робити.

Ймовірно цей спосіб має бути швидшим, оскільки процеси тут не блокують одне одного.
`docker-compose exec counter-service uv run python task6.py`

```
Client 1 started
Client 2 started
Client 0 started
Client 1 finished in 11.62 seconds
Client 2 finished in 11.77 seconds
Client 0 finished in 11.87 seconds
FINAL VALUE: 30000
TOTAL EXECUTION TIME: 11.91 seconds
```

Результат рівно 30000 і, як і передбачувалось, час 11.91 секунд, що майже на 11 секунд менше від песимістичного блокування.

```
hz-node-1-1 | 2026-02-25 20:45:05,124 [ INFO] [hz.zealous_galois.generic-operation.thread-3] [c.h.c.i.p.t.AuthenticationMessageTask]: [hz-node-1]:5701 [dev] [5.6.0] Received auth from Connection[id=74, /172.18.0.2:5701->/172.18.0.3:47628, qualifier=null, endpoint=[172.18.0.3]:47628, remoteUid=2a8175b9-ad1c-430c-87b0-42ca591f5c01, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: 2a8175b9-ad1c-430c-87b0-42ca591f5c01, client name: hz.client_0, client version: 5.6.0
hz-node-1-1 | 2026-02-25 20:45:05,123 [ INFO] [hz.zealous_galois.priority-generic-operation.thread-0] [c.h.c.i.p.t.t.AuthenticationMessageTask]: [hz-node-1]:5701 [dev] [5.6.0] Received auth from Connection[id=73, /172.18.0.2:5701->/172.18.0.3:47612, qualifier=null, endpoint=[172.18.0.3]:47612, remoteUid=e91ac844-ca1c-4619-a551-618319b04954, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: e91ac844-ca1c-4619-a551-618319b04954, client name: hz.client_0, client version: 5.6.0
hz-node-2-1 | 2026-02-25 20:45:05,124 [ INFO] [hz.priceless_sanderson.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [hz-node-2]:5701 [dev] [5.6.0] Received auth from Connection[id=57, /172.18.0.6:5701->/172.18.0.3:34070, qualifier=null, endpoint=[172.18.0.3]:34070, remoteUid=06586c97-426e-4dc1-b4d5-29140ea60d66, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: 06586c97-426e-4dc1-b4d5-29140ea60d66, client name: hz.client_0, client version: 5.6.0
hz-node-2-1 | 2026-02-25 20:45:05,130 [ INFO] [hz.priceless_sanderson.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [hz-node-2]:5701 [dev] [5.6.0] Received auth from Connection[id=58, /172.18.0.6:5701->/172.18.0.3:34074, qualifier=null, endpoint=[172.18.0.3]:34074, remoteUid=2a8175b9-ad1c-430c-87b0-42ca591f5c01, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: 2a8175b9-ad1c-430c-87b0-42ca591f5c01, client name: hz.client_0, client version: 5.6.0
hz-node-2-1 | 2026-02-25 20:45:05,130 [ INFO] [hz.priceless_sanderson.generic-operation.thread-3] [c.h.c.i.p.t.AuthenticationMessageTask]: [hz-node-2]:5701 [dev] [5.6.0] Received auth from Connection[id=59, /172.18.0.6:5701->/172.18.0.3:3406, qualifier=null, endpoint=[172.18.0.3]:3406, remoteUid=e91ac844-ca1c-4619-a551-618319b04954, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: e91ac844-ca1c-4619-a551-618319b04954, client name: hz.client_0, client version: 5.6.0
```

```
hz-node-3-1 | 2026-02-25 20:45:16,925 [ INFO] [hz.serene_euclid.IO.thread-in-2] [c.h.i.s.t.TcpServerConnection]: [hz-node-3]:5701 [dev] [5.6.0] Connection[id=57, /172.18.0.5:5701->software_architecture-counter-service-1.software_architecture_default/172.18.0.3:35024, qualifier=null, endpoint=[172.18.0.3]:35024, remoteUid=2a8175b9-ad1c-430c-87b0-42ca591f5c01, alive=false, connectionType=PYH, planeIndex=-1] closed. Reason: Connection closed by the other side
hz-node-1-1 | 2026-02-25 20:45:16,925 [ INFO] [hz.zealous_galois.IO.thread-in-2] [c.h.i.s.t.TcpServerConnection]: [hz-node-1]:5701 [dev] [5.6.0] Connection[id=74, /172.18.0.2:5701->software_architecture-counter-service-1.software_architecture_default/172.18.0.3:47628, qualifier=null, endpoint=[172.18.0.3]:47628, remoteUid=2a8175b9-ad1c-430c-87b0-42ca591f5c01, alive=false, connectionType=PYH, planeIndex=-1] closed. Reason: Connection closed by the other side
hz-node-2-1 | 2026-02-25 20:45:16,925 [ INFO] [hz.priceless_sanderson.IO.thread-in-0] [c.h.i.s.t.TcpServerConnection]: [hz-node-2]:5701 [dev] [5.6.0] Connection[id=58, /172.18.0.6:5701->software_architecture-counter-service-1.software_architecture_default/172.18.0.3:34074, qualifier=null, endpoint=[172.18.0.3]:34074, remoteUid=2a8175b9-ad1c-430c-87b0-42ca591f5c01, alive=false, connectionType=PYH, planeIndex=-1] closed. Reason: Connection closed by the other side
```

Крок 7. Порівняйте результати кожного з запусків

- Логи до минулих трьох пунктів лише показують, що клієнти успішно підключились, відключились та з'єднання було закрито.
- Як і було вказано в кроці 4 дані було втрачено. Загальна сума становила лише 13375 замість 30000. Причина: race conditions.
- І песимістичний і оптимістичний варіанти блокування видали правильний результат 30000, оскільки гонити даних більше немає в двох випадках.
- Оптимістичний підхід швидший на 11 секунд, причина - песимістичний варіант блокує роботу інших двох процесів, що не мають доступу до ресурсу. Тобто додатковий час йде на очікування звільнення цього ресурсу.

Крок 8. Робота з Bounded queue

Код цього завдання всередині файлу task8.py. Для налаштування Bounded queue на 10 елементів у файлі hazelcast.yml додала такі рядки:

```
queue:
  bounded_queue:
    max-size: 10
    backup-count: 1
```

Клієнти запуснені як 3 процеси, де 1 додає елемент до черги, а інші 2 читають.

Повідомлення читаються одразу. Процеси, що читають, розуміють, що в чергу більше не будуть додаватись повідомлення за допомогою poison pill.

```
docker-compose exec counter-service uv run python task8.py
```

```
Producing 1
Inserted 1
Producing 2
Consumer 2 consumed: 1
Inserted 2
Producing 3
Consumer 1 consumed: 2
Inserted 3
Producing 4
Consumer 2 consumed: 3
Inserted 4
Producing 5
Consumer 1 consumed: 4
Inserted 5
Producing 6
Consumer 2 consumed: 5
Inserted 6
Producing 7
Consumer 1 consumed: 6
Inserted 7
Producing 8
Consumer 2 consumed: 7
Inserted 8
Producing 9
Consumer 1 consumed: 8
```

...

Consumer 2 consumed: 95
Inserted 95
Producing 96
Inserted 96
Consumer 1 consumed: 96
Producing 97
Inserted 97
Producing 98
Consumer 2 consumed: 97
Inserted 98
Producing 99
Consumer 1 consumed: 98
Inserted 99
Producing 100
Consumer 2 consumed: 99
Inserted 100
Consumer 1 consumed: 100
Consumer 2 finished
Producer finished
Consumer 1 finished
All processes finished

Вище я додала частину з усіх логів, на початку роботи та у кінці. Producing - виводиться до того як процес кладе значення у чергу і Inserted - після того як він вже поклав. Можна побачити, що хоча 2 клієнти читають з черги, вони не дублюють роботу один одного, а розділяють її.

Тобто, коли наприклад в черзі з'явилося число 5, обидва клієнти відправляють запит на сервер Hazelcast через метод take(). Сервер отримує запити та віддає елемент тому клієнту, чий запит прийшов першим. Як тільки клієнт 2 забрав 5, Hazelcast видаляє це число з черги. Для Consumer 1 черга стає порожньою, і він продовжує чекати на наступне число. Операція видачі елементів тут атомарна.

Також, якщо продюсер заповнив чергу на 10 елементів і ніхто не викликав take(), то Hazelcast блокує метод put() для нього до моменту звільнення місця. Для перевірки цього я запустила процес, що заповнює чергу, а процеси, що читають починають роботу через 20 секунд.

Part 2, maxed queue
Producer started, waiting for 20 sec...
Producing 1
Inserted 1
Producing 2
Inserted 2
Producing 3
Inserted 3
Producing 4
Inserted 4
Producing 5
Inserted 5
Producing 6
Inserted 6
Producing 7
Inserted 7
Producing 8
Inserted 8

Producing 9
Inserted 9
Producing 10
Inserted 10
Producing 11

Логи зупинилися на значенні 10 до початку роботи консюмерів.

Тут вони продовжили:

Consumers starting now!
Inserted 11
Producing 12
Consumer 1 consumed: 1
Consumer 2 consumed: 2
Inserted 12
Producing 13
Consumer 1 consumed: 3
Consumer 2 consumed: 4
Inserted 13
Producing 14
Consumer 1 consumed: 5
Consumer 2 consumed: 6
Inserted 14
Producing 15
Consumer 1 consumed: 7
Inserted 15
Producing 16
Consumer 2 consumed: 8
Consumer 1 consumed: 9
...

```
hz-node-3-1 | 2026-02-26 18:57:33,988 [ INFO] [hz.vibrant_blackwell.priority-generic-operation.thread-0] [c.h.c.i.p.t
.AuthenticationMessageTask]: [hz-node-3]:5701 [dev] [5.6.0] Received auth from Connection[id=24, /172.18.0.7:5701->/172.18.0
.2:60862, qualifier=null, endpoint=[172.18.0.2]:60862, remoteUuid=ad26cb5e-f817-414c-8be0-c124633f8179, alive=true, connecti
onType=PYH, planeIndex=-1], successfully authenticated, clientUuid: ad26cb5e-f817-414c-8be0-c124633f8179, client name: hz.cl
ient_0, client version: 5.6.0
hz-node-3-1 | 2026-02-26 18:57:33,988 [ INFO] [hz.vibrant_blackwell.generic-operation.thread-0] [c.h.c.i.p.t.Authenti
cationMessageTask]: [hz-node-3]:5701 [dev] [5.6.0] Received auth from Connection[id=23, /172.18.0.7:5701->/172.18.0.2:60846,
qualifier=null, endpoint=[172.18.0.2]:60846, remoteUuid=d0b6586b-c4c0-4272-b146-2d8e5fb5c2fe, alive=true, connectionType=PY
H, planeIndex=-1], successfully authenticated, clientUuid: d0b6586b-c4c0-4272-b146-2d8e5fb5c2fe, client name: hz.client_0, c
lient version: 5.6.0
hz-node-2-1 | 2026-02-26 18:57:33,988 [ INFO] [hz.musing_satoshi.priority-generic-operation.thread-0] [c.h.c.i.p.t.Au
thenticationMessageTask]: [hz-node-2]:5701 [dev] [5.6.0] Received auth from Connection[id=23, /172.18.0.5:5701->/172.18.0.2:
43988, qualifier=null, endpoint=[172.18.0.2]:43988, remoteUuid=bbc5e5dc-0cca-43a7-b997-3e2771d8e2d9, alive=true, connectionT
ype=PYH, planeIndex=-1], successfully authenticated, clientUuid: bbc5e5dc-0cca-43a7-b997-3e2771d8e2d9, client name: hz.clie
nt_0, client version: 5.6.0
hz-node-1-1 | 2026-02-26 18:57:33,994 [ INFO] [hz.happy_faraday.priority-generic-operation.thread-0] [c.h.c.i.p.t.Aut
henticationMessageTask]: [hz-node-1]:5701 [dev] [5.6.0] Received auth from Connection[id=23, /172.18.0.4:5701->/172.18.0.2:5
6114, qualifier=null, endpoint=[172.18.0.2]:56114, remoteUuid=ad26cb5e-f817-414c-8be0-c124633f8179, alive=true, connectionTy
pe=PYH, planeIndex=-1], successfully authenticated, clientUuid: ad26cb5e-f817-414c-8be0-c124633f8179, client name: hz.client
_0, client version: 5.6.0
hz-node-1-1 | 2026-02-26 18:57:33,994 [ INFO] [hz.happy_faraday.generic-operation.thread-1] [c.h.c.i.p.t.Authenticati
onMessageTask]: [hz-node-1]:5701 [dev] [5.6.0] Received auth from Connection[id=25, /172.18.0.4:5701->/172.18.0.2:56118, qua
lifier=null, endpoint=[172.18.0.2]:56118, remoteUuid=d0b6586b-c4c0-4272-b146-2d8e5fb5c2fe, alive=true, connectionType=PYH, p
laneIndex=-1], successfully authenticated, clientUuid: d0b6586b-c4c0-4272-b146-2d8e5fb5c2fe, client name: hz.client_0, clien
t version: 5.6.0
hz-node-1-1 | 2026-02-26 18:57:33,994 [ INFO] [hz.happy_faraday.generic-operation.thread-0] [c.h.c.i.p.t.Authenticati
onMessageTask]: [hz-node-1]:5701 [dev] [5.6.0] Received auth from Connection[id=24, /172.18.0.4:5701->/172.18.0.2:56112, qua
lifier=null, endpoint=[172.18.0.2]:56112, remoteUuid=bbc5e5dc-0cca-43a7-b997-3e2771d8e2d9, alive=true, connectionType=PYH, p
laneIndex=-1], successfully authenticated, clientUuid: bbc5e5dc-0cca-43a7-b997-3e2771d8e2d9, client name: hz.client_0, clien
t version: 5.6.0
```

Також тут додала логи, але вони знову ж таки показують лише успішне підключення клієнтів.